VISUAL GROUNDING × IDENTITY-CONSISTENT REASONING

RoboPIN

Grounded Embodied Reasoning via Pinned Chain-of-Thought

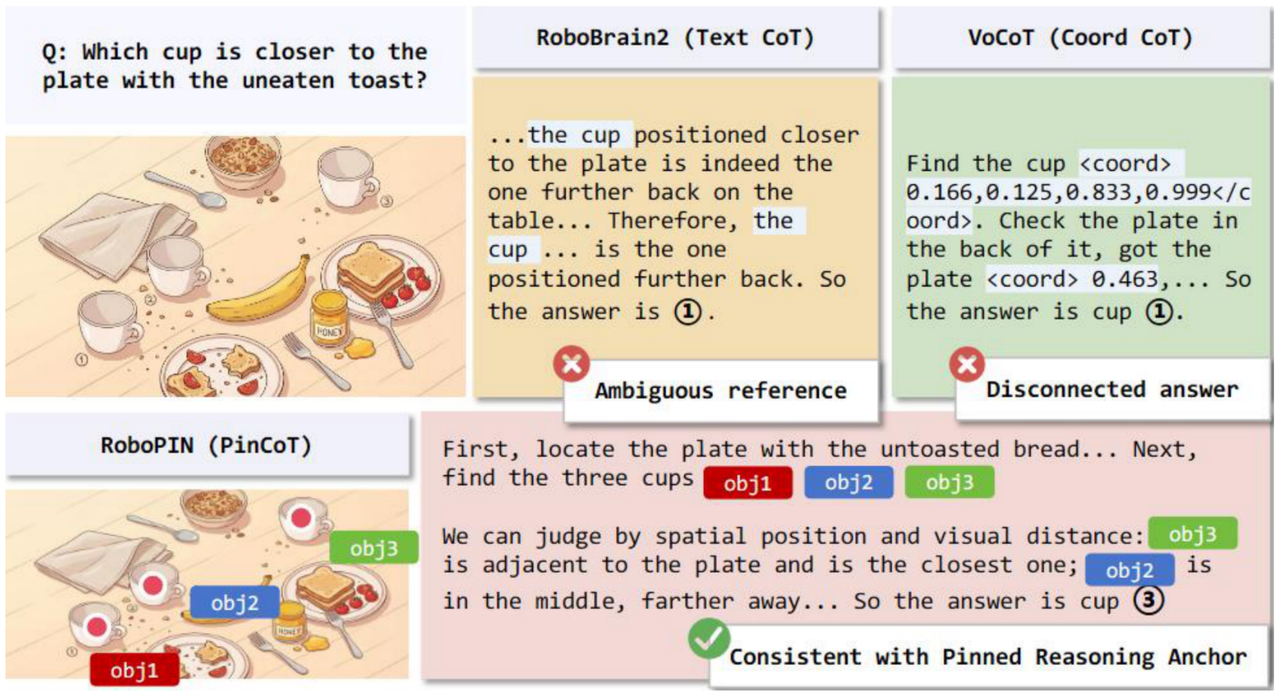
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Tianjin University | ACM Multimedia 2026 · Rio de Janeiro, Brazil

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4B	170K	14	+35.4 pp
model parameters	PinCoT reasoning samples	evaluation benchmarks	CrossPoint vs. Mimo-7B

01 One entity. One persistent identity.



Ground reasoning in persistent visual anchors.

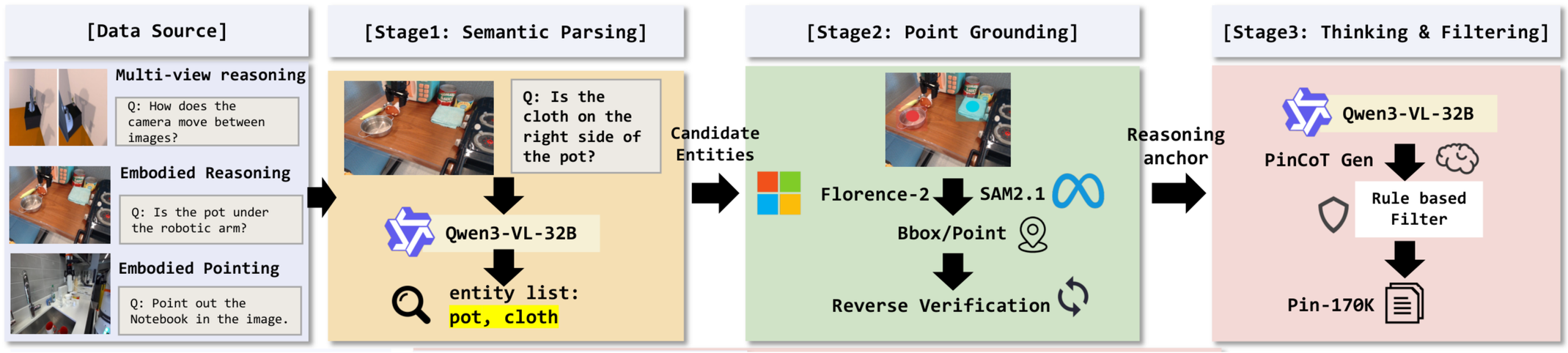
Text CoT
Implicit entity references can drift.

Coordinate CoT
Spatial coordinates alone do not preserve identity.

PinCoT / ours
Bind every step to an explicit, trackable entity.

Name + ID + view + point
Original comparison: paper Figure 1 (left).

02 From data to RoboPIN



Three-stage post-training

SFT	→	CoT-SFT	→	RFT / GRPO
Embodied knowledge		Structured reasoning		Process alignment

03 Results & process analysis

RoboPIN 4B vs. Mimo-Embodied 7B

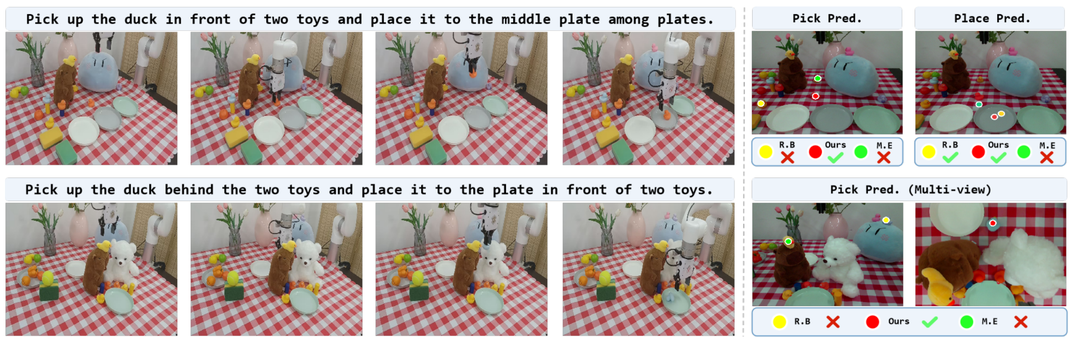
Benchmark	7B	4B / ours	Δ pp
EmbSpatial	76.2	**83.1**	**+6.9**
RoboSpatial	61.8	**66.6**	**+4.8**
CrossPoint	36.8	**72.2**	**+35.4**
BLINK · multi-view	26.3	**72.9**	**+46.6**
RoboRefit	82.3	**89.0**	**+6.7**
RoboAfford	69.8	**72.8**	**+3.0**

Selected results, Tables 1–2; Δ in percentage points.

Cross-step ID coverage: 18.8% → 99.9%
Without RFT → RoboPIN · EmbSpatial, Table 7

04 Zero-shot robot transfer

xArm · static + wrist RGB cameras



Successful executions / 20 per task

20 / 20	14 / 20	11 / 20
Spatial	Multi-view	Long-horizon

Paper Figure 3 and Table 3

Ground every step. Preserve every identity.

PinCoT makes visual evidence trackable across reasoning steps and viewpoints, supporting reliable embodied decisions.

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